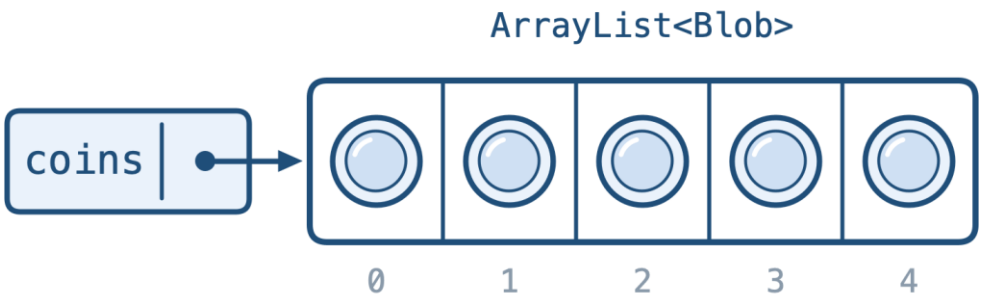
Bonus D - Lists and Loops

One coin is a variable. Twenty coins is a list.



Three coins, all in one spot. Walk the list forward, and you collect two of them.

```
for (int i = 0; i < coins.size(); i++) {
```

```
after frame 1:  score = 2, coins left = 1
```

Look at your game from Chapter 6 or 7. Each coin is a variable, a `new Blob(...)` line, a `move()`, a `drawSprite()`, and an `if`. Adding one more coin takes five pieces in four places. Twenty coins is about a hundred lines.

A hundred coins is about five hundred. `coin1`, `coin2`, `coin3`, typed one at a time, all the way up to `coin100`, until your fingers file a complaint. And somewhere in those five hundred lines is a `coin73` where you meant `coin74`. Good luck finding it.

This chapter fixes that with a **list**: one variable that holds as many coins as you give it, and loops that do the same thing to every one of them.

It's for the student who wants lots of coins. Take it during Chapter 7, or any time after Chapter 6. It takes one meeting. If you're working on your Chapter 7 game, save a copy first, and do this chapter on the copy.

This chapter does it in three moves.

Part	The idea
One	**A bag.** An `ArrayList`, one variable for many things.
Two	**Two loops that both say `for`.** One counts, to make things. One visits, to move and draw them.
Three	**Taking things out.** Removing a coin you collected, safely.

Today's example. Before you start, open this chapter's finished code: **File → Examples**, then **Contributed Libraries → SpriteGame → Lessons → BonusD_Lists**. Run it, and twelve coins and a red hazard drift around a stand-in character. Collect every coin, and a new wave arrives. Keep its window open, and when your sketch does something strange, compare the two. Your own work goes in your own sketch. Appendix 1, section A1.6, walks through opening an example one click at a time.

Ask Sparky. This lesson's Sparky opens from the button on the Bonus D page on Canvas. Good things to ask it: "what is the difference between a counting for loop and a for-each loop?", "my sketch crashes when I collect a coin," "why do we loop backwards?", or "how do I put a list into my Lesson 7 game?" If something won't run, paste in your `setup()` or `playGame()`.

Part One - A bag

D.1 ArrayList

Think of a list as a **bag**. You keep putting things in, and it holds as many as you give it. There are three things you do with a list:

```
ArrayList<Blob> coins;                  // 1. declare it (what kind of bag)
coins = new ArrayList<Blob>();          // 2. create it (get an empty bag)
coins.add(new Blob(...));               // 3. add to it (put something in)
```

> **Aside:** `ArrayList<Blob>`
>
> An `ArrayList` is a list that grows as you add to it. The `<Blob>` says what kind of bag it is: a bag of Blobs. `coins.add(...)` puts one more thing in the bag.

Step 1 - An empty bag

Build it. Open your Chapter 6 sketch. At the top, replace `Blob coin;` with:

```
ArrayList<Blob> coins;      // a list that holds Blobs
```

In `setup()`, replace the line that makes your coin with:

```
  coins = new ArrayList<Blob>();   // an empty list
```

> **Watch out.** Forget that line, and there's no bag. The sketch stops with `NullPointerException` the first time you try to put something in it.

Part Two - Two loops that both say for

Java uses the same word, `for`, for two loops that feel completely different. One **counts**, and you use it to make things. The other **visits**, and you use it to do something to everything already in the list. Keeping them straight is the real skill in this chapter.

Step 2 - Fill it: the counting loop

Build it. In `setup()`, right after the empty list, fill it:

```
  // Counting loop - runs 12 times to create 12 coins
  for (int i = 0; i < 12; i++) {
    coins.add(new Blob(
      this,
      random(width),          // random x position
      random(height),         // random y position
      random(8, 18),          // random radius
      random(-2, 2),          // random horizontal speed
      random(-2, 2),          // random vertical speed
      color(255, 220, 50)     // gold
    ));
  }
```

> **Aside: the counting loop**
>
> `for (int i = 0; i < 12; i++)` says *do this 12 times*. `int i = 0` starts a counter at zero. `i < 12` keeps going while the counter is under 12. `i++` adds one to the counter each time around. So the braces run with `i` at 0, 1, 2, and on up to 11: twelve times.

Run it.

> **Check.** Nothing new on the screen.

The bag has twelve coins in it now. Nothing draws them.

Step 3 - Draw them: the for-each loop

Build it. Where you used to move and draw your coin, **visit** every coin in the bag. Delete your old coin's collision `if` for now.

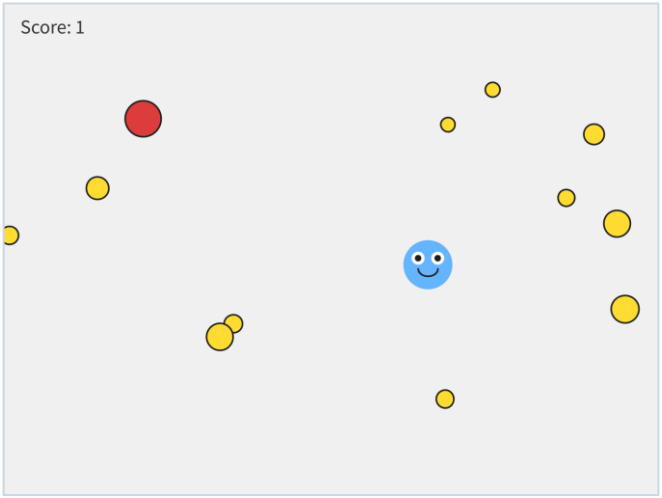
```
  for (Blob c : coins) {
    c.move();
    c.drawSprite();
  }
```

Aside: the for-each loop

`for (Blob c : coins)` reads *for each Blob `c` in `coins`*. It hands you the coins one at a time, calling each one `c` while it's your turn with it, and runs the braces once for every coin in the list.

Run it.

Check. The world is full of coins.



Twelve coins, from one loop.

Watch out. The bag is a bag. It can't draw itself. `coins.drawSprite();` stops the sketch with:

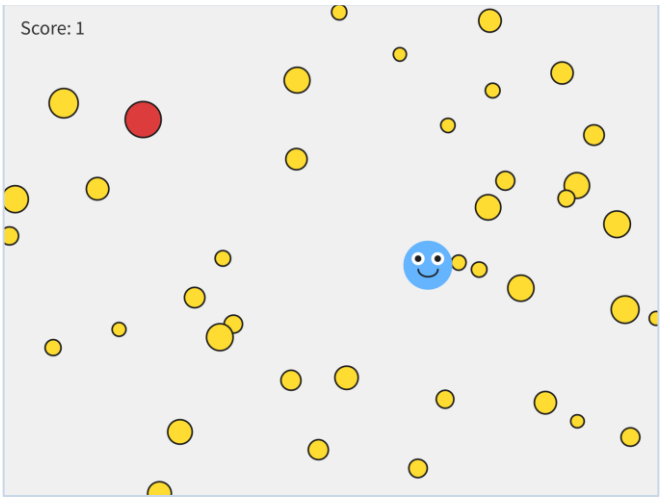
```
The function drawSprite() does not exist.
```

Use the for-each loop to reach into the bag and ask each coin.

Watch out. Mix the two loops up, and the messages get strange. `for (int i = 0; i < coins; i++)` gets `The operator < is undefined for the argument type(s) int, ArrayList<Blob>`, because you can't count up to a bag. `for (Blob c : 12)` gets `Can only iterate over an array or an instance of java.lang.Iterable`, because you can't visit a number. One counts. One visits.

Step 4 - How many?

Build it. Change the `12` in the counting loop to `40`, and run. Then try `3`. How many feels right?



Forty. Same five lines of code.

> **Watch out.** Put the counting loop in `draw()` by mistake, and it adds twelve coins every frame. After one second there are about 700, and the sketch grinds to a halt. Making things goes in `setup()`.

Part Three - Taking things out

D.5 The problem with removing

With a list, collecting a coin means taking it **out** of the list. Removing something from a list *while you're walking through it* causes a real problem.

Picture five coins in a row. You're standing on coin 2, and you take it out. Every coin after it shifts back one place to fill the gap. Then the loop steps you forward to place 3, and the coin that *was* at 3 is at 2 now, behind you. You just skipped it.

It's not just a picture. Put three coins right on top of your player and walk the list forward, removing each one you touch: you collect two of them, and one gets skipped.

The fix is to walk **backwards**, from the end of the list toward the front. Taking a coin out then only shifts coins you've already passed, so every coin gets its turn. You don't have to work out why. Recognize it as *the safe way to remove things while looping*, and copy it when you need it.

Step 5 - Walk backwards

Build it. Replace your for-each loop with the backwards loop, with no collision check yet:

```
// Backwards loop - the safe way to remove things while looping
for (int i = coins.size() - 1; i >= 0; i--) {
  Blob c = coins.get(i);       // grab the coin at position i
  c.move();
  c.drawSprite();
}
```

Run it.

Check. Nothing looks different. The two loops do the same thing until you take something out.

Aside: `size()`, `get()`, and counting from 0

`coins.size()` is how many things are in the list right now. Every item has a number for its place, called its **index**, and a list starts counting at **0**. So twelve coins are numbered 0 to 11, and the last one is `coins.size() - 1`. `coins.get(i)` hands you the coin at place `i`. And `i--` counts **down**, one at a time.

Step 6 - Collect

Build it. Inside the backwards loop, after the coin is drawn, add the collision:

```
    if (player.collidesWith(c)) {
      coins.remove(i);           // take it out of the list
      score = score + 1;
    }
```

Run it, and collect a few.

Check. Touch a coin, it's gone, and the score goes up.

Watch out. Take a coin out inside a **for-each** loop, and the sketch stops with `ConcurrentModificationException`. That's a big word for *you changed the list while you were walking through it*. The backwards loop is the fix.

Watch out. Start the backwards loop at `coins.size()`, without the `- 1`, or count up with `i++` where `i--` belongs, and the sketch stops with:

```
IndexOutOfBoundsException: Index 12 out of bounds for length 12
```

Twelve coins are numbered 0 to 11, so there's no coin 12 to get.

Step 7 - A new wave

`coins.size()` shrinks every time you collect one. When it hits zero, the board is clear, and you get to decide what that means.

Build it. After the backwards loop, send a fresh wave:

```
  // An empty list means the board is clear:  send a new wave
  if (coins.size() == 0) {
    for (int i = 0; i < 12; i++) {
      coins.add(new Blob(this, random(width), random(height),
                         random(8, 18), random(-2, 2), random(-2, 2),
                         color(255, 220, 50)));
    }
  }
```

Clear the board, and watch it fill up again. Or set `playerWon = true;` in there, and the game ends. Your call.

Step 8 - A coin on a click

Build it. Add a coin wherever you click:

```
void mousePressed() {
  coins.add(new Blob(this, mouseX, mouseY,
                     random(8, 18), random(-2, 2), random(-2, 2),
                     color(255, 220, 50)));
}
```

Step 9 - A second list

Your turn. Make your hazards a list too: `ArrayList<Blob> hazards;`, filled with a counting loop, and visited with a **for-each** loop. You never take a hazard out, so the for-each loop is all you need.

D.7 Make it yours

Your turn. Change how many coins are in a wave, how big they are, and how fast.

Your turn. Optional: any of these.

- Make each wave bigger than the last. Count the waves in a variable, and use it in the counting loop.
- Add one more hazard to the hazards list every time a wave clears.
- Put a list into your Chapter 7 game. Convert your coins, and keep the hazards as they are. In `restartGame()`, `coins = new ArrayList<Blob>();` and then the counting loop give you a fresh, full board every restart: the same rebuild-from-scratch idea as the rest of `restartGame()`.
- Make a collected coin **split**: add two smaller coins to the list in its place.

Chapter summary

You built	It's an example of
`ArrayList<Blob> coins;`	a list: one variable for many things
`coins = new ArrayList<Blob>();`	an empty bag
`for (int i = 0; i < 12; i++) { coins.add(...); }`	the **counting** loop, to make things
`for (Blob c : coins) { ... }`	the **for-each** loop, to visit things
the backwards loop, with `coins.remove(i)`	the safe way to take things out while looping
`if (coins.size() == 0)`	an empty list, and a new wave

New syntax in this chapter

Syntax	Where you met it
`ArrayList<Blob>`, `new ArrayList<Blob>()`	D.1, Step 1
`add()`	D.1, Step 2
`for (int i = 0; i < 12; i++) { }`, `i++`	Step 2
`for (Blob c : coins) { }`	Step 3
`size()`, `get(i)`, `i--`	Step 5
`remove(i)`	Step 6

Vocabulary

`ArrayList` -- a list that holds many objects of the same kind, and grows as you add more. (D.1)

counting loop -- a loop that repeats a set number of times. Used to **make** things. (Step 2)

for-each loop -- a loop that visits every item already in a list, one at a time. Used to **visit** things. (Step 3)

index -- an item's place in a list. Lists start counting at 0. (Step 5)

`size()` -- how many items are in the list right now. (Step 5)

Check your understanding

1. What's the difference between the counting loop and the for-each loop? When do you use each one?
2. Why doesn't `coins.drawSprite();` work?
3. Twelve coins. What's the index of the last one?
4. Why do we loop backwards when we take things out?
5. Your sketch stops with `ConcurrentModificationException`. What did you do?